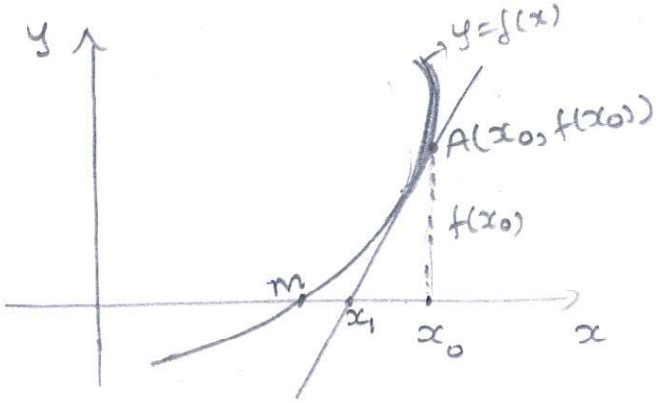
Newton-Raphson method

Geometrical interpretation of the method.



Since m is a root of the equation $f(x)=0$ near the value $x=x_0$, the curve $y=f(x)$ crosses the x-axis at the point $x=m$ which lies near the point $x=x_0$.

The equation of the tangent to the curve at the point $A(x_0, f(x_0))$ is

$$\frac{y-f(x_0)}{x-x_0} = f'(x_0) \quad \rightarrow (1)$$

At the point x_1, where this tangent meets x-axis, we have $y=0$ and equation (1) yields

$$\frac{-f(x_0)}{x-x_0} = f'(x_0)$$

on simplification

$$\boxed{x = x_0 - \frac{f(x_0)}{f'(x_0)}} \rightarrow 1^{st} \text{ approximation}$$

The generalized formula for Newton Raphson method is

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

Newton-Raphson method has a second order or quadratic convergence.

① Find the cube root of 24, correct to three places of decimal by Newton-Raphson method.

Soln:- Finding the cube root of 24 is same as solving the equation of $x^3 - 24 = 0$

Let $f(x) = x^3 - 24$

$f'(x) = 3x^2$

we have Newton-Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^3 - 24}{3x_n^2}$$

$$x_{n+1} = \frac{2x_n^3 + 24}{3x_n^2} \rightarrow (1)$$

Now, for $n=0$, $x_0 = 2.8$, we have

$$x_1 = \frac{2(2.8)^3 + 24}{3(2.8)^2}$$

$$= 2.887$$

Second approximation, put $n=1$, $x_1 = 2.887$ in (1)

$$x_2 = \frac{2(2.887)^3 + 24}{3(2.887)^2}$$

$$= 2.8845$$

Third approximation, put $n=2$, $x_2 = 2.8845$ in (1)

$$x_3 = \frac{2(2.8845)^3 + 24}{3(2.8845)}$$

$$= 2.8844$$

Thus the cube root of 24 is 2.884.

(2) Using N-R method, find the root near 2.9 of the equation $x + \log_{10} x = 3.375$ correct to four decimal places.

$$\log_{10} x = \frac{\log_e x}{\log_e 10}$$

Soln: $f(x) = x + \log_{10} x - 3.375$

$$f'(x) = 1 + \frac{1}{x} \log_{10} e$$

$$\log_{10} e = \frac{\log_e e}{\log_e 10}$$

By N-R method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{(x_n + \log_{10} x_n - 3.375)}{1 + \frac{1}{x_n} \log_{10} e} \rightarrow (1)$$

Now, for $n=0$, $x_0 = 2.9$

$$x_1 = 2.9 - \frac{2.9 + \log_{10}(2.9) - 3.375}{1 + \frac{1}{2.9} \log_{10} e}$$

$$= 2.9 - \frac{0.0126}{1.1497}$$

$$= 2.9109$$

For second approximation, put $n=1$, $x_1 = 2.9109$ in (1)

$$x_2 = 2.9109 - \frac{2.9109 + \log_{10}(2.9109) - 3.375}{1 + \frac{1}{2.9109} \log_{10} e}$$

$$= 2.9109 - \frac{0.0001}{1.1492}$$

$$= 2.91099.$$

The approximated root is 2.9109

(3) Find the root of the equation $\tan x = x$, near $x = 4.5$ correct to four decimal places.

Soln: Let $f(x) = \tan x - x$

$$f'(x) = \sec^2 x - 1$$
$$= \tan^2 x$$

By Newton-Raphson method, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{\tan x_n - x_n}{\tan^2 x_n} \rightarrow (1)$$

First approximation, put $n = 0$, $x_0 = 4.5$, we get

$$x_1 = 4.5 - \frac{\tan(4.5) - 4.5}{\tan^2(4.5)}$$

$$= 4.4936$$

Second approximation, put $n = 1$, $x_1 = 4.4936$ in (1)

$$x_2 = 4.4936 - \frac{\tan(4.4936) - 4.4936}{\tan^2(4.4936)}$$

$$= 4.4934$$

For third approximation put $n = 2$, $x_2 = 4.4934$ in (1)

$$x_3 = 4.4934 - \frac{\tan(4.4934) - 4.4934}{\tan^2(4.4934)}$$

$$= 4.4934$$

As x_2 and x_3 are identical, we take the approximate root as 4.4934.

(4) Find a root of $3x - \cos x = 1$ correct to four decimal places by using Newton-Raphson method.

Soln: Let $f(x) = 3x - \cos x - 1$

We have $f(0) = -2$ and $f(1) = 1.45970$

∴ The root lies between 0 and 1. Hence we take the initial guess x_0 as 0.7

By Given eqn $f(x) = 3x - \cos x - 1$

$$f'(x) = 3 + \sin x$$

∴ By Newton-Raphson method, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{3x_n - \cos x_n - 1}{3 + \sin x_n} \rightarrow (1)$$

First approximation, put $n=0$, $x_0 = 0.7$ in (1)

we get $x_1 = 0.6080$

Second approximation, put $n=1$, $x_1 = 0.6080$ in (1)

∴ $x_2 = 0.6071$

Third approximation, put $n=2$, $x_2 = 0.6071$ in (1)

∴ $x_3 = 0.6071$

Hence the approximate root is 0.6071.

(5) Find a real root of the equation $x = e^{-x}$, using Newton-Raphson method by taking initial guess $x = 1$.

Soln:. we write the equation in the form $xe^x = 1$

$$\Rightarrow \quad f(x) = xe^x - 1$$

$$f'(x) = xe^x + e^x$$

∴ By Newton-Raphson method, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n e^{x_n} - 1}{(x_n e^{x_n} + e^{x_n})} \quad \rightarrow (1)$$

Put $n = 0$ and $x_0 = 1$ in (1) for the first approximation

$$\therefore x_1 = 1 - \frac{e^1 - 1}{(e + e)}$$

$$= 0.6839$$

For the second approximation, put $n = 1$, $x_1 = 0.6839$ in (1)

$$\therefore x_2 = 0.5774$$

For the third approximation, put $n = 2$, $x_2 = 0.5774$ in (1)

$$\therefore x_3 = 0.5672$$

For the fourth approximation, put $n = 3$, $x_3 = 0.5672$ in (1)

$$x_4 = 0.5671$$

Hence the approximate root is 0.5671.

⑥ Using Newton-Raphson method, find the real root of the equation $x\sin x + \cos x = 0$ near $x = \pi$.

Soln:- Let $f(x) = x\sin x + \cos x$

$$f'(x) = x\cos x + \sin x - \sin x$$

$$= x\cos x$$

∴ By Newton-Raphson method we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n} \rightarrow (1)$$

1st approximation, put $n=0$, $x_0 = \pi$, in (1)

$$x_1 = 2.8232$$

2nd approximation, put $n=1$, $x_1 = 2.8232$ in (1)

$$x_2 = 2.7985$$

3rd approximation, put $n=2$, $x_2 = 2.7985$ in (1)

$$x_3 = 2.7983$$

Hence the approximate root is 2.7983.